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Classification of (micro)plastics using cathodoluminescence and machine learning

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ABSTRACT

Microplastics are a growing environmental and toxicological concern, having been found in the remotest locations of the earth and within multiple organs of the human body. However, the scale of the problem is not yet fully known as the smallest micro- and nanoplastics (MNPs) cannot accurately be measured due to limitations in measurement and detection techniques. In this paper, we combine the nanoscale resolution of Scanning Electron Microscopy (SEM) with the spectroscopic power of cathodoluminescence (CL) to show that six of the most common plastics – HDPE, LDPE, PP, PA, PS and PET – have unique spectra that could help in identifying the smallest MNPs. This was done by building a spectral database using 111 plastic samples from reference and consumer plastics with different sizes (0.001–1 mm), colours (*e.g.* black, blue, green, red, white/transparent and yellow) and shapes (*e.g.* irregular, fibre, spheres). We then trained multiple classification models using an Artificial Neural Network approach. With the use of these classification models we were able to classify the six plastics, including difficult samples such as black coloured plastics, based on their CL spectra with 97% accuracy, showing that our approach is robust towards sample differences. As most misclassifications occurred between LDPE and HDPE, a separate model for LDPE and HDPE allowed for >99% accuracy for the classification of HDPE and LDPE by using a two-step approach. This novel "proof-of-concept" to MNP analysis demonstrates the utility of SEM paired with CL to characterise microplastics with detailed spatial and chemical resolution warranting its further development and adoption.

1. Introduction

Plastics have become seemingly indispensable in our daily lives owing to their versatility, weight and cost. Between 1950 and 2015, annual plastics production grew from 2 Mt to 381 Mt [1], an increase of nearly 200 times, as they found uses in applications as diverse as packaging, construction and textiles [2]. However, our reliance on plastics has led to the generation of an estimated 6300 Mt of plastic waste, with only an estimated 9% being recycled [1]. This plastic waste is likely to pose serious risks to both wildlife and humans with large pieces of plastic causing birds and marine animals to become trapped and die. [3] Whilst there is still much uncertainty surrounding the persistence and lifetimes of plastics in the environment [4,5], plastic waste has been shown to disintegrate due to, for example, UV exposure, thermal exposure, mechanical friction and microbial attack in the environment [6,7]. These processes can result in the release of micro- and nanoplastics (MNPs) [8], small plastic particles of which the health risks and the environmental impact are still uncertain [9–11]. These particles are suspected of causing serious adverse health effects as their small size allows for them to be taken up by various organisms, including humans. In addition, these MNPs can also act as a "trojan horse", carrying different types of organic pollutants and toxic metals on its surface [12]. Although there is serious ground for concern surrounding this developing environmental pollutant, the health risks associated with (micro)plastic pollution are still not fully understood [13,14]. Whatever the risk, with plastic waste generation expected to grow to 155–265 Mt per year by 2060, micro- and nanoplastics presence in the environment and accompanied risks will only grow [15].

To effectively tackle MNPs released from plastic waste, and understand the potential health risks they pose, we need to understand their presence in the environment. For this, accurate detection techniques that give information about their size, number and distribution in the environment are required. Furthermore, to develop suitable mitigation strategies we also need to understand the sources of MNPs indicating the need to chemically identify different types of MNP particles at single

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Abbreviations

AFM	Atomic Force Microscopy
AI	Artificial Intelligence
ANN	Artificial Neural Networks
CL	Cathodoluminescence
EDX	Energy Dispersive X-ray Spectroscopy
EM	Electron Microscopy
FTIR	Fourier Transform Infrared Spectroscopy
HDPE	High-density Poly Ethylene
LDPE	Low-density Poly Ethylene
LIBS	Laser-induced Breakdown Spectroscopy
LoD	Limit of Detection
LOF	Local Outlier Factor Score
ML	Machine Learning
MNPs	Micro- and Nanoplastics
PA	Polyamide
PC	Principle Component(s)
PCA	Principle Component Analysis
PE	Polyethylene
PET	Polyethylene Terephthalate
PP	Polypropylene
PS	Polystyrene
SEM	Scanning Electron Microscopy
XRF	X-ray Fluorescence

particle level.

The detection and identification of plastics is a great challenge and becomes even greater as the particles get smaller [16]. Chemical composition of the smallest fraction of microplastics (the nanoplastics) can be obtained through various methods, with thermal analytical based techniques being the most promising. However, as this is a bulk-based technique, it doesn't provide all of the information needed to fully identify these small particles. It is widely accepted that using a combination of different analysis techniques that allows for the identification of mass distributions (via thermal methods), physical (*e.g.* size and shape) and chemical properties (*e.g.* polymer type) is optimal, as combining them leads to complete characterisation of the pollutant [17]. Identification of the latter, physical and chemical properties, is commonly achieved by combining microscopic and spectroscopic techniques. For example, optical microscopy is a quick and easy method for determining quantity, size and shape of plastics, however this gives no chemical information so there is also a high possibility of misidentification [18]. By combining optical microscopy with vibrational spectroscopy techniques such as Fourier Transform Infrared spectroscopy (FTIR) or Raman (or both), chemical information can be acquired that allows for positive identification of plastic types [19–21]. However, these techniques come with their own disadvantages, with detection limits of approximately 20 μm and 1 μm for μ-FTIR and μ-Raman respectively and both techniques suffering from problems due to additives and colourants, especially with black samples as it absorbs all the light which is needed for analysis [19,22]. Recently, IR and Raman have been combined with other microscopic techniques such as atomic force microscopy (AFM) and SEM, respectively. This has extended their spatial resolution range and allowed for the positive identification of MNPs <1 μm [23,24].

Scanning electron microscopy (SEM) is a promising technique for plastic detection as it can probe a range from tens of nanometres up to millimetres, thus encompassing both the nanoplastic (tens of nanometres to 1 μm) and microplastic (1 μm–5 mm) size regimes [19,25]. SEM also provides morphological information and is commonly coupled with energy-dispersive x-ray analysis (EDX) which allows for elemental analysis on the same length scale. SEM-EDX has been used in the analysis of other environmental pollutants such as asbestos [26,27] and particulate matter [28,29]. However, whilst it is useful for distinguishing plastics from contaminants, as most plastics are hydrocarbon polymers consisting of C and H (and occasionally O and Cl), SEM-EDX is often limited in plastic type identification.

Cathodoluminescence (CL) is the emission of light which is generated when a material is excited with an electron beam. To obtain the best spatial resolution, CL signal is generally produced inside a SEM. By equipping an electron microscope with a cathodoluminescence detector, materials are excited through the electron beam inside the microscope, so the main source of energy is the kinetic energy of the electrons. As this energy is relatively high, this excitation will result in energy loss through a variety of processes among which fluorescence, recombination-induced luminescence and chemiluminescence are examples. CL is light that is emitted in a wide spectral range ranging from the ultraviolet through the mid infrared and is dependent on the optical properties of the underlying material. This luminescence is affected by factors such as composition and microstructure [30] and has been used to investigate defect-induced degradation in laser diodes [31] and trace elements in gemstones [32]. SEM-CL is a technique that is most commonly used to characterise semiconductors [33,34] and geological samples [35,36]. In theory, the analysis spot size is limited by that of the electron beam, allowing for CL spectral analysis at the same resolution as that of the electron microscope used, thus allowing for the analysis of particles in the micro- and nano size fraction. As this technique investigates electronic composition and microstructure as opposed to elemental composition (like EDX) or bonding (FTIR and Raman), it offers a unique new perspective for the identification and characterisation of plastics. This can be done by, for example, lowering the size limit for identification or following the chemical changes of a plastic during degradation.

It has been shown that many types of plastics are cathodoluminescent and produce unique spectra that may be used for identification [37–40]. SEM based CL imaging as technique to study polymer materials and microplastics have also been previously suggested for its potentially high resolution [41]. However, the spectral differences between some plastics (*e.g.* HDPE and LDPE) are low, and with the spectral quality expected to decrease with decreasing plastic size, correct classification will be challenging. This is compounded by the fact that the origins of most CL peaks for plastics are not yet fully understood. Some plastics have been investigated in more detail. For example, through comparison with electroluminescence it has been suggested that in the CL spectrum for PP, the peak at 328 nm is due to the natural fluorescence of the material, the peak at 415 nm is due to chemiluminescence and the peak at 505 nm due to charge recombination [38,42]. However, the main peak in many of the samples at 573 nm, as well as the peaks in most other plastics, remains unidentified. To overcome these challenges, artificial intelligence (AI) classification algorithms can be used. With the growing pace of advances in data analytics, AI and machine learning (ML) approaches are increasingly being used in combination with spectroscopic techniques for the identification of explosives [43], battery life [44], food [45] and even Martian rocks [46]. With a supervised machine learning algorithm it is possible to learn the mapping function from the input variables (features of the model) to the output variable.

In this proof-of-concept study, we trained machine learning classifiers that learn to predict the type of plastics given the CL spectra of those plastics. ML algorithms have been used in combination with numerous analytical techniques for the identification of microplastics, including FTIR [47–49], Laser-induced breakdown spectroscopy (LIBS) [50] and X-ray fluorescence (XRF) [51]. However, to the best of our knowledge, AI has never been used in conjunction with CL. Using a mix

of micro- and macroscale plastics from both reference and consumer sources, we aim to produce a database of CL spectra for six of the most abundant plastic types. Using this database, we will investigate the potential of CL for plastic identification in combination with AI classification algorithms. We hope that the approach pioneered in this work demonstrates the power of AI in combination with EM-based CL spectroscopy for the efficient detection and identification of plastics. In future, full utilisation of the resolution of electron microscopy (EM) may allow this to be extended to the smallest MNPs that provide the biggest challenges.

2. Materials and methods

2.1. Materials

The plastic types studied in this work were high-density polyethylene (HDPE), low-density polyethylene (LDPE), polystyrene (PS), polypropylene (PP), polyamide (PA) and polyethylene terephthalate (PET). For each of the plastic types, approximately 20 different samples were collected (HDPE = 14, LDPE = 18, PA = 17, PET = 22, PS = 18, PP = 22) with a total of 111 samples. In order to include the influence of additives and colours and make the model more robust, a mixture of reference and consumer plastics were collected. A figure displaying a selection of the plastic material (70 samples) used in this work along with a table with a detailed description per sample can be found in the Supporting Information (see Fig. S1 and Table 1).

2.2. SEM sample preparation

Samples were prepared for SEM analysis by either cutting cross-sections or by filtering suspensions. The cross sections of the plastics were cut with a razor blade followed by transferring them onto an aluminium SEM-stub covered with carbon coated tape (JEOL Europe BV). The suspensions were filtered under reduced pressure (200 mbar) over 25 mm gold coated polycarbonate filters with pore sizes of 0.8 μm and 0.1 μm (TJ Environmental). To avoid the agglomeration of the particles, a small amount (~15 μL per sample) of TWEEN-20 (Sigma-Aldrich) was added to the suspension. After filtration, the filter was transferred onto an aluminium SEM-stub covered with a carbon coated tape (JEOL Europe BV). The samples were coated with a thin conductive film of carbon using a carbon evaporator (Quorum Q150 T carbon evaporator) to make the samples electronically conductive. Fig. S3 in the Supplementary Information illustrates the stepwise workflow of preparing and measuring cross-sections from large plastics.

2.3. SEM-CL set-up and measurements

Cathodoluminescence (CL) experiments were performed inside a Tescan MAIA III GMH field emission scanning electron microscope. The CL spectra were collected and measured using a Delmic SPARC-Spectral system equipped with a high resolution CCD camera (Andor Newton DU920P) mounted on a SPARC ATM spectrograph. The light was collected with an aluminium paraboloid mirror with an acceptance angle of 1.49π. The spectrograph contains a motorized entrance slit and a turret with two different gratings.

CL spectra were recorded using the hyperspectral imaging mode with a VIS-NIR optical module optimized for measurements in the λ = 400–1000 nm range and a spectral resolution of 0.83 nm/pixel. The measurements were performed using a 300 lines per mm ruled grating, blazed for a free space wavelength at λ = 500 nm with a resolution of 0.96 nm and centred at λ = 550 nm. To excite the plastics, a beam energy of 15 kV was set to the material. The entrance slit in front of the spectrograph was closed between 200 μm and 400 μm. Typical integration times of 200 ms/pixel and 400 ms/pixel were applied. Hardware binning of the CCD pixels (full vertical binning and 2 times horizontally) was applied to improve the signal-to-noise ratio (SNR). Background spectra, i.e., without electron beam current, and (if present) cosmic rays were subtracted from the data. To capture any variation within the sample, each sample was measured in at least four different representative 7 × 7 μm regions to ensure potential fillers and additives are included in the measurements. The resulting spectrum from the 7 × 7 μm area is an aggregate from 121 individual pixel spectra, with each pixel having a dimension of 636 × 636 nm. With these settings, it is possible to measure spectra of particles in the sub-micron size range. The raw, background-corrected spectra of all samples in this work are provided in .csv in the Supporting Information.

Feature Selection:

In order to find suitable spectral features by which the different plastics can be identified, the raw CL spectra were aggregated for each plastic. That is, for each of the plastics the median intensity at each wavelength was calculated in order to make one aggregated spectrum per plastic type. After aggregating all the spectra, features in the spectra were found by applying the following three methods;

(i) Peak finding on the aggregated spectra. Using R package "pracma" [52] with the following rule-set:
   - Minimum number of increasing/decreasing steps before a peak is reached = 1 (for PA we used 2)
   - Minimum (absolute) height of a peak = 0
   - Minimum peak distance i.e. the minimum distance (in indices) peaks have to be counted: 10 for HDPE, LDPE, PA and PS, 30 for PET, 50 for PP

(ii) Peak finding on the derivatives plots: Similar to the first method, the R package "pracma" was applied with the same rule-set, but now the derivative plots of the aggregated spectra were used. For this, the derivatives of the lines between 10 consecutive points were calculated. To only use positive numbers, the absolute value of the minimum derivative was added.

(iii) Finding anomalies in the aggregated spectra using the "Local Outlier Factor Score (LOF)" [53] method. In this method, points with an LOF score of significantly larger than 1, are likely to be outliers. As a local neighbourhood, two nearest neighbours of a point were considered. In order to determine the outliers, the top 2 percentiles of the LOF scores was set as a threshold.

By applying these three methods, a total of 212 features were found for all of the plastic types. As an example the selected features for PS are shown in Fig. 1 (Features PS). These points are then used as potential input for the classification model.

2.4. Machine learning models

First, principle component analysis (PCA) was used on the selected features in order to reduce the dimensionality of the data input for the classification model. The data was centred and scaled for PCA implementation. Artificial Neural Networks (ANN) were then used for the

Table 1
Accuracy of the five-fold cross-validation of the three models.

Fold Number	1	2	3	4	5	Mean	Standard Deviation
General Model	95.29	97.80	91.25	90.54	93.75	93.73	0.03
Combined Model	97.65	96.70	97.50	97.30	100.00	97.83	0.01
PE Model	100.00	100.00	100.00	95.83	96.55	98.48	0.02

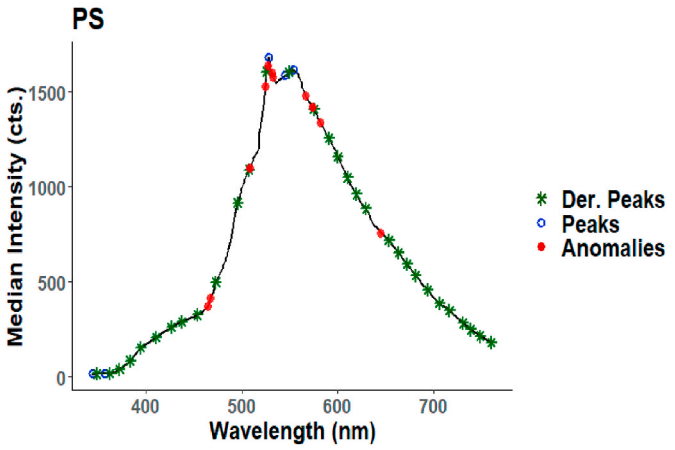


Fig. 1. Features selected for PS using peaks, peaks in the derivatives and anomalies.

classification of plastics. We first developed a model using all plastics (General Model). Then, as LDPE and HDPE had very similar CL-spectra, two more classification models were developed: one model with LDPE and HDPE grouped for the classification of PE, PP, PET, PA, and PS (Combined Model) and one for the classification of LDPE and HDPE (PE-model).

The dataset was randomly split into 80% for the training set and 20% for the test set. A 5-fold cross validation was performed on the training set for hyperparameter tuning of the three models. The test set was used to check the performance of the models [54–56]. Since we had multiple measurements per sample, we made sure that all the measurements for each sample were always either in the training set or in the validations/test set.

The models were developed using the "H2O package" [57] in the programming language "R" [58]. The H2O package uses multi-layer feedforward artificial neural networks trained with stochastic gradient descent using back-propagation [59].

3. Results and discussion

3.1. CL spectra and its features

CL spectra of six of the most common plastics types were collected in the wavelength range from 300 nm to 800 nm from 111 different samples. The plastics that were investigated all had different origins, ranging from reference samples that were bought directly from chemical suppliers and received from plastic producers, and samples that were obtained through our own material collection. The CL spectra of the six plastic types (HDPE, LDPE, PA, PET, PP and PS) are visualised in Fig. 2. In this figure, CL spectra of both white/transparent and black coloured plastics are shown with varying intensities, showing the range between the lowest and highest measured intensities of black and white/transparent plastics.

Similar to what has been previously demonstrated in the work of Pakzad et al. [40], the spectra of the plastic types have visible differences, such as spectral shape, intensities and maximum peak values. As the sample set consisted of plastics from different sources, there was great variance in composition between samples of the same plastic type, *e.g.* in used colourants and additives and the presence of possible contaminants. Interestingly, no spectral differences are observed when comparing the spectra of black coloured samples to the spectra of white/transparent samples of the same plastic type, as can be seen in Fig. 2. The CL spectra of different coloured PS samples are shown in Fig. S2 in the Supporting Info. As we can see in both Fig. 2 and Fig. S2, there is some intensity variation observed between samples, however this seems to be independent of colour. The intensity can also vary significantly depending on the accuracy of the alignment of the parabolic mirror for light collection in the CL system. As plastic materials do not generate that much cathodoluminescence signal inside a SEM as most conventionally used materials (*e.g.* rock or minerals), a slightly off alignment contributes significant to the measured intensity. These results indicate that the SEM-CL technique is suitable for the detection of a wide range of plastics, including difficult samples such as black coloured plastics or samples with different additives which hinder standard micro-spectroscopic techniques such as μ-FTIR and μ-Raman.

3.2. PCA results

A feature selection approach on the CL spectra of the six plastic types led to a set of 212 spectral features that were often highly correlated. In order to decrease the dimensionality of the dataset, the features were subjected to PCA. Accuracy tests on the training set were performed with varying numbers of principal components (PCs) to select the number of PCs that would be used in the classification models. That is, the model was first run using 2 PCs only and then one PC was added to the model in each run followed by calculating the average accuracy of the cross-validation folds. When there was no improvement in the average accuracy for two consecutive runs, then the number of PCs with the highest accuracy was chosen as the number of PCs for that model. 10, 9 and 12 PCs were selected for the General Model, the Combined Model and the PE model respectively.

The cumulative variance explained per principle component is shown in Fig. 3 and shows that the first 2 PCs describe over 94% of the variance and that if 7 PCs are used almost 100% (99.92%) is described. Furthermore, in the score plot of PC1 and PC2 in Fig. 3 it is seen that the scores of HDPE and LDPE are highly correlated with each other and are mainly described by PC1. PP is also mainly described by PC1 and is to a lower extent correlated with HDPE and LDPE. The scores of PET and PA are highly correlated with each other and appear orthogonally to the PE

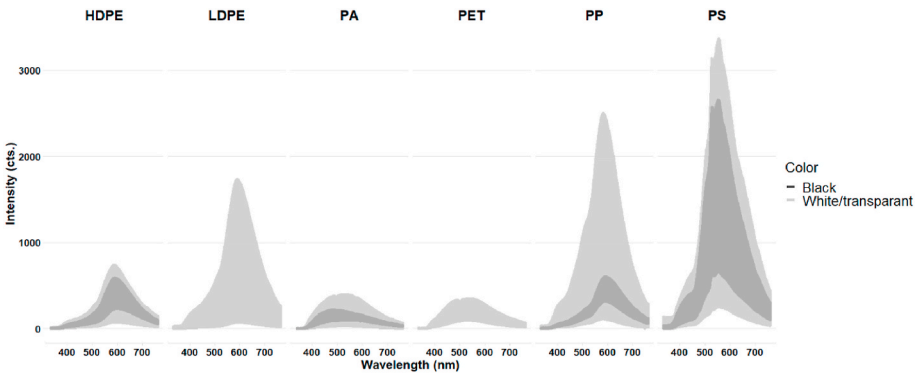


Fig. 2. CL-spectra of the plastics used in the classification model (aggregated using median values for each type of plastic).

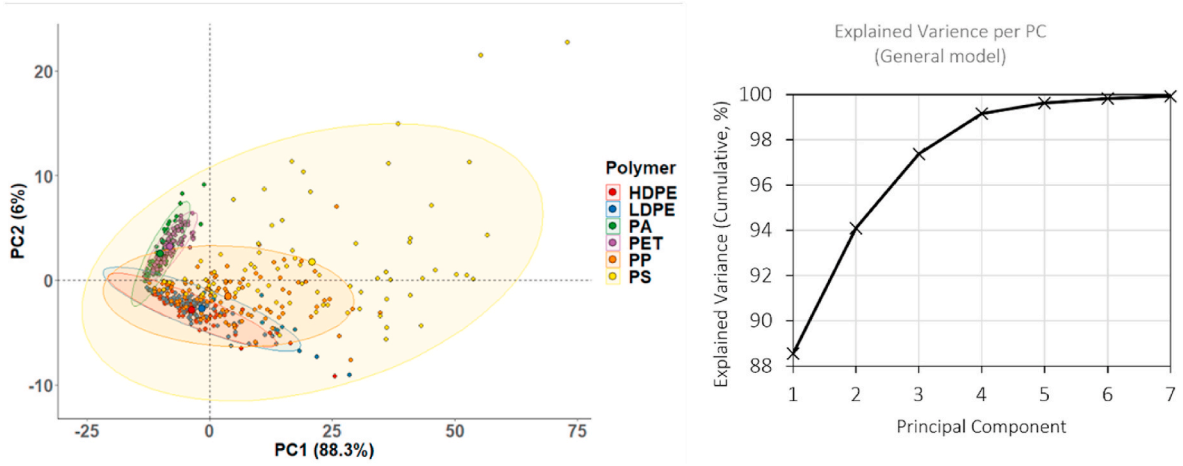


Fig. 3. Score plot for the General model (left) and the explained variance per PC (right). Performance of the Models.

clusters, mainly being described by PC2. These observation are supported by looking at the loading plots as shown in Fig. S4 in the Supporting Information. PC1 shows positive loadings between peaks at ~400, 500 and 600 nm which describe the spectrums of HDPE, LDPE and PP as shown in Fig. 2 of the manuscript. The PC2 loadings show a strong positive contribution in the region of the spectrum from ~380 to 550 nm which is where the signal from PET and PA is the strongest. PC2 also shows strong negative loadings between 550 and 750 nm, the region of the spectrum which describes HDPE and LDPE.

In the CL spectra of both PA and PET, spectral features can be observed in the wavelength range between 400 and 500 nm. It is known that the 415 nm peak in CL spectra of PP originates from the chemiluminescence of excited carbonyl radicals (C═O*) [38]. As both PET and PA contain a carbonyl (C═O) as functional group, the similarity in CL spectra could be attributed to the similarity of this functional group, as it is present in PET, PA as well as oxidized PP. The score plots for the Combined model and the PE model have been provided in Fig. S5 and Fig. S6 of the SI.

Table 1 provides the results on the performance of the models after cross-validation. The General model, the Combined model and the PE model had an average accuracy of 94%, 98% and 98% respectively. The low standard deviations in the cross-validation results show that the models are reliable.

In Fig. 4 the confusion matrices for the test sets are shown. Here, it can be seen that the models produced similar or better results on the test sets compared to the results of the cross-validation. The General model had an overall accuracy of 94% (Fig. 4a). As expected - given the overlap of the clusters made by principle component analysis - the misclassifications occurred mostly between LDPE and HDPE and between PET and PA. As measurements were performed on the submicron scale (with each pixel of 636 nm giving a spectrum), this model seems promising for the identification and classification of plastics that are too small for other conventional measurement techniques such as μ-FTIR and μ-Raman. In the GP model (Fig. 4b), all samples were classified correctly except for 3 PET samples that were misclassified as PA. The model had therefore a very high accuracy on the test set (97%).

Fig. 4c presents the confusion matrix for the PE model. Here, it can be seen that the PE model predicted both of the classes correctly, producing a perfect result on the test set. Given the very similar chemical structure of HDPE and LDPE, their consequent similar spectra (see Fig. 2) and the almost indistinct clusters from the PCA (see Fig. S6) the high accuracy of this model shows the strength of the developed ANN model combined with spectroscopy for plastic identification. This is beneficial for research focused on finding and identifying potential sources of microplastics in the environment. By distinguishing LDPE and HDPE from each other, the sources of different grades of PE (HDPE and LDPE) can be identified, allowing the design of effective mitigation strategies.

By splitting the General model into a two-step model, there was a limited improvement in overall accuracy from 94% to 97%, however this not reflected for all polymers as PET was less accurately classified. However, with these different models we have shown how the technique can be adapted to the research question of the user. If distinction between HDPE and LDPE is essential, then either the general model or the 2-step model can be used. If this is not the case, the combined model can

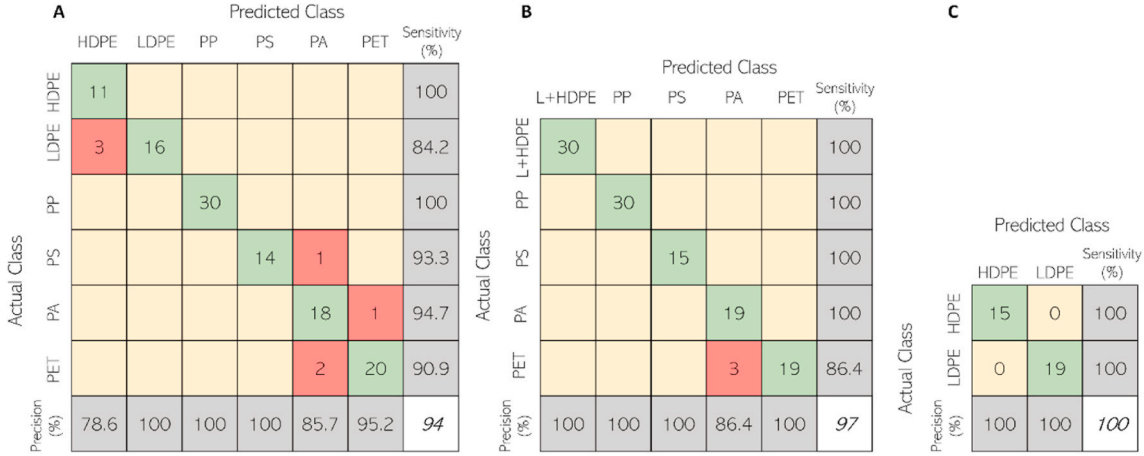


Fig. 4. Confusion matrices on the test set for the General model (A), the Combined model (B) and the PE model (C).

be used. So, in the end, optimisation of the model can be tuned according to the desired research question.

With this work, we have demonstrated that the AI supported SEM-CL technique is suited for the classification of plastics, as it is able to distinguish between PE, PET, PS, PA and PP and even between HDPE and LDPE with high accuracy. By building a spectral database consisting of reference and consumer plastics (0.001–1 mm) and training a two-step Artificial Neural Network classification model, we were able to classify the six plastics with an accuracy of over 97%. The colour of the samples didn't have an impact on the spectral quality, indicating that the SEM-CL technique is highly applicable for environmental samples that have a complex variety of plastics with different pollutants, colours, additives, etc. As current detection techniques don't allow for single particle analysis in the whole MNPs size region, the SEM-CL technique in combination with AI offers a novel approach to gain more insight into the sources of both micro- and nanoplastics in our surroundings. The knowledge gained on the sources of MNPs will significantly contribute to designing suitable mitigation strategies that will effectively tackle this prominent environmental pollutant. Furthermore, the SEM-CL technique enables investigation of physical aspects such as particle size, polymer type, aspect ratio and surface area. These properties are expected to influence the related health effects of the particles and need to be investigated to gain more knowledge of the risk they pose. This work shows that the SEM-CL technique is promising for the detection and identification of MNPs, still research needs to be done to investigate how weathering and particle size influence the CL signal in order to apply the technique on real environmental samples (including weathered MNPs) and define a limit of detection (LoD). As we move into the sub-micron size region, the interaction volume of the electron beam may exceed the size of the particles measured, thus potentially resulting in additional CL emission from the underlying material. It is therefore crucial for the SEM imaging parameters and CL imaging conditions (*e.g.* beam intensity, CL dwell times) to be optimised with regards to measuring particles of different sizes and materials. Furthermore, we want to encourage further development of the technique by, for example, automation to help decrease measurement times and costs. The use of this technique is currently limited as SEM facilities with CL are scarcer in comparison with other techniques such as SEM-EDX and performing of the measurements is highly labor intensive. Automation or CL mapping would allow for analysis of larger fields of view to help enable the application of this promising technique for complex environmental samples with MNPs present. This work has shown that the SEM-CL technique is promising for the classification of (micro- and nano)plastics and we hope that in the future this approach can be further developed by expanding the CL-database with spectra of mixed polymers (*e.g.* PET/PA, PE/PP co-polymers), other plastics types (*e.g.* PVC, PC, PMMA) and by demonstrating its applicability on real environmental samples with micro- and nanoplastics present to eventually help us enable a greater understanding of the sources of this environmental pollutant.

4. Conclusions

In conclusion, this proof-of-concept study has demonstrated that the developed classification model(s) using machine learning methods can successfully classify plastics based on their SEM-CL spectra with an accuracy of >97%. As measurements were performed in the submicron size range (121 spectra from a 7 × 7 μm area, pixel size = 636 nm), this model seems promising for the identification and classification of plastics that are too small for other conventional measurement techniques such as μ-FTIR and μ-Raman. Further research will focus on expanding the CL-database (*e.g.* weathered samples, PVC, PC, PMMA) and testing the model performance on CL spectra of micro- and nanoplastics in the submicron size region, if necessary, accompanied by further model optimalisation.

CRediT author statement

Elena M. Höppener: Conceptualization, Investigation, Resources, Writing – Original Draft, Visualization **M. (Sadegh) Shahmohammadi:** Conceptualization, Methodology, Software, Validation, Formal analysis, Data Curation, Writing – Review & Editing, Visualization **Luke Parker:** Conceptualization, Investigation, Resources, Writing – Original Draft, Visualization **Sieger Henke:** Writing – Review & Editing, Supervision, Project administration **Jan Harm Urbanus:** Writing – Review & Editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data generated in this project can be found as a .csv file in the electronic supporting information.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.talanta.2022.123985.

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